

Mia Francis Bonini

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SUMMARY

Engineer with 5+ years of experience in cardiovascular devices, clinical data integration, and finite element modeling. Experienced in collaborating with physicians, engineers, and cross-functional teams to evaluate device performance and patient-specific outcomes. Proven ability to translate complex technical analyses into clinically actionable insights and communicate findings to diverse audiences. Background includes supporting device development, training collaborators, and interfacing between engineering and clinical stakeholders.

EDUCATION

Ph.D. in Biomedical Engineering University of Michigan, Ann Arbor, MI <i>Dissertation Title: Patient-Specific Computational Modeling of Cardiovascular Interventions and Pathologies</i>	Nov. 2025
Master of Science in Biomedical Engineering University of Michigan, Ann Arbor, MI	Apr. 2022
Bachelor of Science in Mechanical Engineering University of Portland, Portland, OR • University of Applied Sciences Upper Austria	May 2020 Jan. – Apr. 2018

RESEARCH EXPERIENCE

University of Michigan, Cardio Biomechanics Lab Research Staff Advisor: David Nordsletten • Lead development of an image-based fluid–structure interaction (FSI) modeling framework for patient-specific simulation of aortic disease and cardiovascular device interventions, enabling in-silico evaluation of treatment strategies and hemodynamic risk factors.	Ann Arbor, MI Nov. 2025 – current
Graduate Student Research Assistant Advisor: David Nordsletten • Collaborated with clinicians and engineers to develop and validate patient-specific cardiovascular models integrating imaging and hemodynamic data, achieving <10% error versus clinical measurements. • Conducted patient-specific CFD studies of aortic valve replacement and valve-in-valve procedures, demonstrating that annular enlargement reduces transvalvular gradients by 92% without increasing blood stasis. • Assessed valve repair strategies in patients with left ventricular assist devices, translating simulation results into clinically relevant insights on blood stasis, valve function, and right ventricle loading. • Performed in-silico experiments to optimize fenestration size and placement for a novel aortic dissection device, informing surgical design decisions and supporting a \$415k funded device development effort. • Interfaced regularly with cardiac surgeons and radiologists to interpret imaging, discuss procedural strategies, and contextualize modeling results • Organized and led meetings with engineers, cardiac surgeons, radiologists, and students to ensure project completion, leading to 5 first-author articles, 7 co-authored articles, 2 book chapters, and 7+ presentations at national and international conferences.	Aug. 2020 – Nov. 2025

Wake-Forest University, Department of Biomedical Engineering
Undergraduate Researcher

Winston-Salem, NC
Jun – Aug. 2019

- Developed an automated imaging analysis pipeline to quantify long-term spaceflight-induced changes in spinal muscle morphology and composition, reducing manual processing time and improving reproducibility.
- Integrated subject-specific imaging and morphometric data into the finite element Global Human Body Models Consortium (GHBMC) framework to inform biomechanical simulations of spinal loading and musculoskeletal adaptation.

WORK EXPERIENCE

Intel Corporation

Folsom, CA

Engineering Intern – Solid-State Drive & Thermal Solutions

Jun. – Aug. 2018

- Prepared and experimentally evaluated solid-state drives (SSDs) with integrated thermal management solutions in wind tunnel and still-air chamber environments, analyzing temperature profiles to optimize device thermal performance and reliability.
- Standardized laboratory testing protocols and documentation to reduce operator variability, improve reproducibility, and enhance workflow reliability across mechanical and thermal validation processes.

HONORS AND AWARDS

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| 1. Functional Imaging and Modeling of the Heart Early-Career Scholarship Award, \$645 | Jun. 2025 |
| 2. Marvin Kirsh Resident Research Symposium, 1 st Place Basic Science, \$300 | Mar. 2025 |
| 3. University of Michigan Rackham Conference Travel Grant, \$1500 (x5) | 2021-2025 |
| 4. U.S. Association for Computational Mechanics Travel Award, \$775 | Jul. 2024 |
| 5. University of Michigan Rackham Graduate Student Research Grant, \$3000 | Nov. 2023 |
| 6. EUROMECH Young Researcher Award | Jul. 2022 |

ADDITIONAL EXPERIENCE

- Trained and mentored 14 undergraduate and MS students in all stages of cardiovascular computational modeling, enhancing technical proficiency and research capabilities.
- Guided 3 students in preparing and presenting research at the Summer Biomechanics, Bioengineering, and Biotransport Conference, increasing their professional exposure.
- Led a student through the process of writing a research paper, resulting in a co-authored publication
- Planned and led community engagement events for the department's graduate students
- Mentored two prospective PhD students from non-traditional paths or historically disadvantaged backgrounds during the PhD application process

SKILLS & INTERESTS

- **Technical:** finite element analysis (FEA), computational fluid dynamics (CFD), fluid-structure interaction (FSI), hemodynamic modeling of medical devices, in-silico valve deployment, in-silico experimentation, medical image segmentation, simulations, statistics, scientific writing and communication
- **Clinical & Imaging:** CT/MRI/echocardiography data analysis, 3D reconstruction, medical image segmentation, cardiovascular physiology
- **Image Processing:** 3D Slicer, Materialize Mimics, ITK Snap, Eidolon, OsiriX
- **Computer:** MATLAB, Python, Linux Bash, Ansys, COMSOL, Abaqus
- **Personal:** skiing, volleyball, reading, card games, hiking